

Terraform-06

1). Watch the Terraform-06 video.

- Terraform Modules: main.tf file created

```
main.tf \ X main.tf aws_instance variables.  
main.tf > ...  
1 module "dev-webserver" {  
2   source = "./aws_instance"  
3   ami    = "ami-02f3f602d23f1659d"  
4 }  
5
```

- In aws_instance main.tf, variable.tf, outputs.tf files created

```
main.tf \ main.tf aws_instance X variables.tf outputs.tf  
aws_instance > main.tf > resource "aws_instance" "webserver" > abc instance_type  
1 resource "aws_instance" "webserver" {  
2   ami          = var.ami  
3   instance_type = "t3.micro"  
4 }
```

```
main.tf \ main.tf aws_instance variables.tf X outputs.tf  
aws_instance > variables.tf > variable "ami"  
1 variable "ami" {  
2   type = string  
3 }
```

```
main.tf \ main.tf aws_instance variables.tf outputs.tf X  
aws_instance > outputs.tf > output "instance_id"  
1 output "instance_id" {  
2   value = aws_instance.webserver.id  
3 }
```

- Now do terraform init

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform init  
Initializing modules...  
- dev-webserver in aws_instance  
Initializing provider plugins found in the configuration...  
- Finding latest version of hashicorp/aws...  
- Installing hashicorp/aws v6.50.0...  
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)  
  
Initializing the backend...
```

Terraform has created a lock file `.terraform.lock.hcl` to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

- Terraform validate and plan

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform validate
Success! The configuration is valid.

PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform plan

Terraform used the selected providers to generate the following execution plan. Resources to be
+ create

Terraform will perform the following actions:

# module.dev-webserver.aws_instance.webserver will be created
+ resource "aws_instance" "webserver" {
```

- Terraform apply

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform apply

Terraform used the selected providers to generate the following execution plan. Resources to be
+ create

Terraform will perform the following actions:

# module.dev-webserver.aws_instance.webserver will be created
+ resource "aws_instance" "webserver" {
```

```
Enter a value: yes

module.dev-webserver.aws_instance.webserver: Creating...
module.dev-webserver.aws_instance.webserver: Still creating... [00m10s elapsed]
module.dev-webserver.aws_instance.webserver: Creation complete after 18s [id=i-0f500ba88fdf0d160]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

- Now check in instance new ec2 instance created

Successfully initiated termination (deletion) of i-075589ccc60abb25b

Instances (1) Info

Last updated 13 minutes ago

Connect

Instance state

Actions

Saved filter sets

Running

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public
☐		i-0f500ba88fdf0d160	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-13

2). Execute the script shown in the video.

- Terraform Modules: Development created main.tf file

```
main.tf aws_instance X main.tf development X variables.tf outputs.tf
development > main.tf > module "dev-webserver"
1 module "dev-webserver" {
2     source      = "../aws_instance"
3     ami         = "ami-02f3f602d23f1659d"
4
5 }
```

- Now same use aws_instance file like main.tf, variables.tf, outputs.tf

```
main.tf aws_instance X main.tf development variables.tf outputs.tf
aws_instance > main.tf > ...
1 resource "aws_instance" "webserver" {
2     ami           = var.ami
3     instance_type = "t3.micro"
4 }
5
```

```
main.tf aws_instance X main.tf development variables.tf X
aws_instance > variables.tf > variable "ami"
1 variable "ami" {
2     type = string
3 }
```

```
... main.tf aws_instance X main.tf development variables.tf outputs.tf X
aws_instance > outputs.tf > output "instance_id"
1 output "instance_id" {
2     value = aws_instance.webserver.id
3 }
```

- Now do Terraform init, plan, apply

```
PS C:\Users\LENOVO\Desktop\Terraform-basic\development> terraform init
Initializing modules...
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.50.0...
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)

Initializing the backend...
```

```
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\LENOVO\Desktop\Terraform-basic\development> terraform plan
```

```
Plan: 1 to add, 0 to change, 0 to destroy.
```

```
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exact state.
PS C:\Users\LENOVO\Desktop\Terraform-basic\development> terraform apply
```

```
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
+ create
```

```
Enter a value: yes
```

```
module.dev-webserver.aws_instance.webserver: Creating...
module.dev-webserver.aws_instance.webserver: Still creating... [00m10s elapsed]
module.dev-webserver.aws_instance.webserver: Creation complete after 17s [id=i-0362a2984e1191634]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

- New Ec2 instance created

Instances (1) Info							Last updated 1 minute ago	Refresh	Connect	Instance state ▼	Actions
Saved filter sets		Find Instance by attribute or tag (case-sensitive)									
Running ▼											
Instance state = running X		Clear filters ▼									
☐	Name 🔗 ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Zone					
☐		i-0362a2984e1191634	Running 🔍 🔍	t3.micro	3/3 checks passed	us-east-1c					

- Terraform Workspace: main.tf created

```
main.tf X variables.tf
aws_instance C:\Users\LENOVO\Desktop\Terraform-basic\aws_instance\main.tf
1 resource "aws_instance" "webserver" {
2     ami           = var.ami
3     instance_type = "t3.micro"
4 }
5
```

- Now do Terraform workspace new project and check list and default

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform workspace new projectA
Created and switched to workspace "projectA"!
```

```
You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
```

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform workspace list
default
* projectA
```

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform workspace select default
Switched to workspace "default".
PS C:\Users\LENOVO\Desktop\Terraform-basic>
```

3). Provision EC2, S3, and VPC using Terraform modules.

- Create a new project as Terraform Project in that 3 folders created **ec2**, **s3**, **vpc**
- Create **main.tf** file

```
main.tf X
main.tf > module "ec2" > abc instance_type
1 provider "aws" {
2   region = "us-east-1"
3 }
4
5 module "vpc" {
6   source     = "./vpc"
7   vpc_cidr   = "10.0.0.0/16"
8   vpc_name   = "terraform-vpc"
9 }
10
11 module "s3" {
12   source     = "./s3"
13   bucket_name = "arun-sai-bucket-123456"
14 }
15
16 module "ec2" {
17   source       = "./ec2"
18   ami          = "ami-0c02fb55956c7d316"
19   instance_type = "t3.micro"
20   instance_name = "terraform-ec2"
21 }
```

- In **EC2** **main.tf**, **variables.tf**, **output.tf** files created

```
main.tf X
ec2 > main.tf > resource "aws_instance" "server"
1 resource "aws_instance" "server" {
2
3   ami          = var.ami
4   instance_type = var.instance_type
5
6   tags = {
7     Name = var.instance_name
8   }
9 }
```

```
variables.tf X
ec2 > variables.tf > variable "instance_name"
1 variable "ami" {}
2 variable "instance_type" {}
3 variable "instance_name" {}
```

```
outputs.tf X
ec2 > outputs.tf > output "instance_id"
1 output "instance_id" {
2   value = aws_instance.server.id
3 }
```

- Now in **S3 main.tf, variables.tf, outputs.tf** files created

```
main.tf X
s3 > main.tf > resource "aws_s3_bucket" "bucket"
1 resource "aws_s3_bucket" "bucket" {
2   bucket = var.bucket_name
3 }

variables.tf X
s3 > variables.tf > variable "bucket_name"
1 variable "bucket_name" {}

outputs.tf X
s3 > outputs.tf > output "bucket_id"
1 output "bucket_id" {
2   value = aws_s3_bucket.bucket.id
3 }
```

- Now in **Vpc main.tf, variables.tf, outputs.tf** files created

```
main.tf X
vpc > main.tf > resource "aws_vpc" "myvpc"
1 resource "aws_vpc" "myvpc" {
2   cidr_block = var.vpc_cidr
3
4   tags = {
5     Name = var.vpc_name
6   }
7 }

variables.tf X
vpc > variables.tf > variable "vpc_name"
1 variable "vpc_cidr" {}
2 variable "vpc_name" {}

outputs.tf X
vpc > outputs.tf > output "vpc_id"
1 output "vpc_id" {
2   value = aws_vpc.myvpc.id
3 }
```

- Now do Terraform init, plan and apply

```
PS C:\Users\LENOVO\Desktop\Terraform -project> terraform init
Initializing modules...
- s3 in s3
- vpc in vpc
- ec2 in ec2
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.50.0...
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)

Initializing the backend...
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
PS C:\Users\LENOVO\Desktop\Terraform -project> terraform plan
```

- Terraform plan

Terraform will perform the following actions:

```
# module.ec2.aws_instance.server will be created
+ resource "aws_instance" "server" {
  + ami                  = "ami-0c02fb55956c7d316"
  + arn                  = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone     = (known after apply)
```

- Terraform apply

```
PS C:\Users\LENOVO\Desktop\Terraform -project> terraform apply
module.s3.aws_s3_bucket.bucket: Refreshing state... [id=arun-sai-bucket-123456]
module.vpc.aws_vpc.myvpc: Refreshing state... [id=vpc-0d26ec62e114efbc9]
```

Terraform used the selected providers to generate the following execution plan. Resource **+ create**

Terraform will perform the following actions:

```
# module.ec2.aws_instance.server will be created
```

Enter a value: yes

```
module.ec2.aws_instance.server: Creating...
module.ec2.aws_instance.server: Still creating... [00m10s elapsed]
module.ec2.aws_instance.server: Creation complete after 18s [id=i-065f7b018c8b9de49]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

- Resources Created Successfully

- New EC2 instance created

instances (1) [Info](#) less than a minute ago [Connect](#) [Instance state](#)

Saved filter sets
Running [Find Instance by attribute or tag \(case-sensitive\)](#)

Instance state = running [X](#) [Clear filters](#) [v](#)

☐	Name v	Instance ID	Instance state v	Instance type v	Status check	A
☐	terraform-ec2	i-065f7b018c8b9de49	Running v v	t3.micro	3/3 checks passed	u

- S3 bucket Created

[Amazon S3](#) > [Buckets](#) > arun-sai-bucket-123456

amazon S3 <

Buckets

general purpose buckets

directory buckets

arun-sai-bucket-123456 [Info](#)

[Objects](#) Metadata Properties Permissions Metrics Management

- VPC created

Your VPCs

[VPCs](#) VPC encryption controls

Your VPCs (2) [Info](#)

[Find VPCs by attribute or tag](#)

☐	Name	VPC ID	State
☐	-	vpc-096ec8a65d98c6729	Available
☐	terraform-vpc	vpc-0d26ec62e114efbc9	Available

- Now do terraform destroy

```
PS C:\Users\LENOVO\Desktop\Terraform -project> terraform destroy
module.s3.aws_s3_bucket.bucket: Refreshing state... [id=arun-sai-bucket-123456]
module.vpc.aws_vpc.myvpc: Refreshing state... [id=vpc-0d26ec62e114efbc9]
module.ec2.aws_instance.server: Refreshing state... [id=i-065f7b018c8b9de49]
```

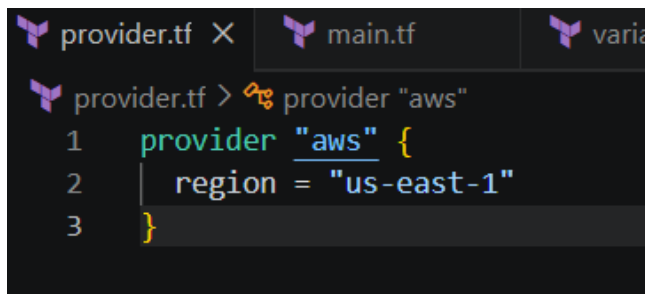
```
Enter a value: yes

module.vpc.aws_vpc.myvpc: Destroying... [id=vpc-0d26ec62e114efbc9]
module.s3.aws_s3_bucket.bucket: Destroying... [id=arun-sai-bucket-123456]
module.ec2.aws_instance.server: Destroying... [id=i-065f7b018c8b9de49]
module.s3.aws_s3_bucket.bucket: Destruction complete after 1s
module.vpc.aws_vpc.myvpc: Destruction complete after 2s
module.ec2.aws_instance.server: Still destroying... [id=i-065f7b018c8b9de49, 00m10s elapsed]
module.ec2.aws_instance.server: Still destroying... [id=i-065f7b018c8b9de49, 00m20s elapsed]
module.ec2.aws_instance.server: Still destroying... [id=i-065f7b018c8b9de49, 00m30s elapsed]
module.ec2.aws_instance.server: Destruction complete after 32s

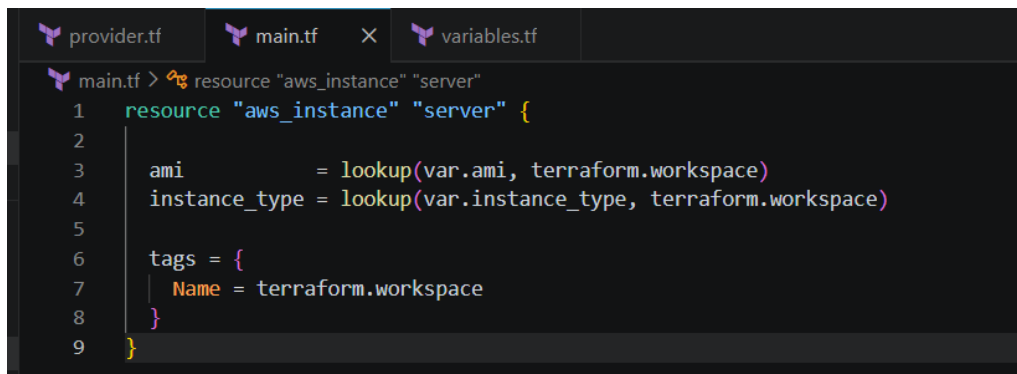
Destroy complete! Resources: 3 destroyed.
```


4). Provision EC2 for 3 different environments (Dev, Staging, and Prod) using Terraform workspaces.

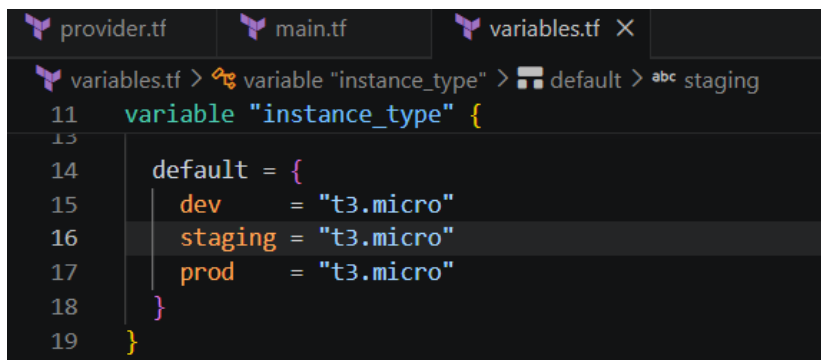
- Create a new folder and create **provider.tf**, **main.tf**, **variables** files



```
provider.tf > provider "aws"
1 provider "aws" {
2   region = "us-east-1"
3 }
```

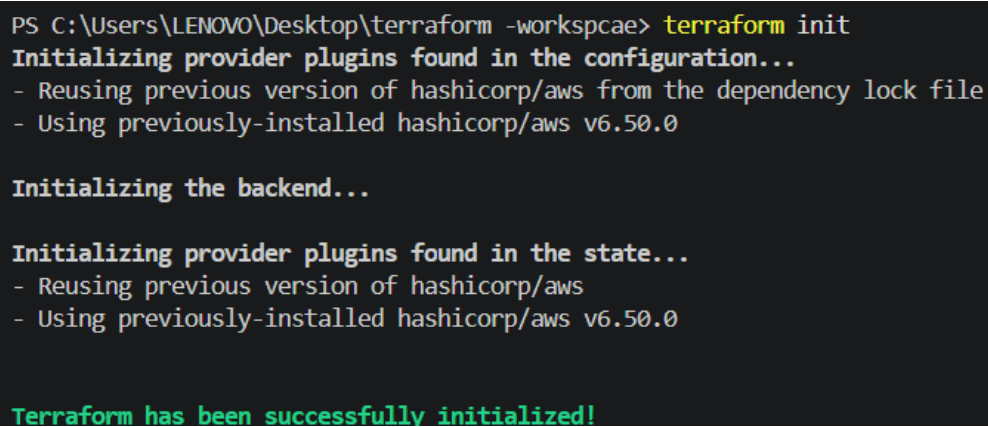


```
main.tf > resource "aws_instance" "server"
1 resource "aws_instance" "server" {
2
3   ami          = lookup(var.ami, terraform.workspace)
4   instance_type = lookup(var.instance_type, terraform.workspace)
5
6   tags = {
7     Name = terraform.workspace
8   }
9 }
```



```
variables.tf > variable "instance_type" > default > staging
11 variable "instance_type" {
12
13   default = {
14     dev    = "t3.micro"
15     staging = "t3.micro"
16     prod   = "t3.micro"
17   }
18 }
19 }
```

- After creating files run command terraform init



```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.50.0

Initializing the backend...

Initializing provider plugins found in the state...
- Reusing previous version of hashicorp/aws
- Using previously-installed hashicorp/aws v6.50.0

Terraform has been successfully initialized!
```

- Run these commands for dev, staging and prod

```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace new dev
Workspace "dev" already exists
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace new staging
Workspace "staging" already exists
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace new prod
Workspace "prod" already exists
```

```
workspace "prod" already exists
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace list
default
dev
* prod
staging
```

- Now switch to dev and do terraform apply command

```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace select dev
Switched to workspace "dev".
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform apply
```

```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform apply
aws_instance.server: Refreshing state... [id=i-084de588533ac9de9]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no changes to be made.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
```

- Now switch to staging and do terraform apply

```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace select staging
Switched to workspace "staging".
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform apply
aws_instance.server: Refreshing state... [id=i-034e11657128a1f26]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place
```

Enter a value: yes

```
aws_instance.server: Modifying... [id=i-034e11657128a1f26]
aws_instance.server: Still modifying... [id=i-034e11657128a1f26, 00m10s elapsed]
aws_instance.server: Still modifying... [id=i-034e11657128a1f26, 00m20s elapsed]
aws_instance.server: Still modifying... [id=i-034e11657128a1f26, 00m30s elapsed]
aws_instance.server: Modifications complete after 37s [id=i-034e11657128a1f26]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.
```

- Now switch to prod and do terraform apply

```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace select prod
Switched to workspace "prod".
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform apply
aws_instance.server: Refreshing state... [id=i-003c6b1f237f628f7]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no changes to make.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
```

- We can check list of terraform workspace created

```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace list
default
dev
* prod
staging

PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace show
prod
```

- In AWS Ec2 instances are created

Instances (3) [Info](#) Last updated 9 minutes ago [Connect](#) [Instance state](#)

Saved filter sets: Running

Instance state = running

☐	Name ↗	Instance ID	Instance state ▼	Instance type ▼	Status check	Avail
☐	prod	i-003c6b1f237f628f7	Running 🔍 🔍	t3.micro	3/3 checks passec	us-e2
☐	staging	i-034e11657128a1f26	Running 🔍 🔍	t3.small	3/3 checks passec	us-e2
☐	dev	i-084de588533ac9de9	Running 🔍 🔍	t3.micro	3/3 checks passec	us-e2

- After created successfully destroy all

```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace select dev
Switched to workspace "dev".
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform destroy
aws_instance.server: Refreshing state... [id=i-084de588533ac9de9]

Terraform used the selected providers to generate the following execution plan.
- destroy

Terraform will perform the following actions:

# aws_instance.server will be destroyed
```

```
Destroy complete! Resources: 1 destroyed.
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace select staging
Switched to workspace "staging".
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform destroy
aws_instance.server: Refreshing state... [id=i-034e11657128a1f26]

Terraform used the selected providers to generate the following execution plan. Resource ac
- destroy
```

```
Destroy complete! Resources: 1 destroyed.
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace select prod
Switched to workspace "prod".
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform destroy
aws_instance.server: Refreshing state... [id=i-003c6b1f237f628f7]

Terraform used the selected providers to generate the following execution plan. Resource actions are
- destroy
```

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources in workspace "prod"?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_instance.server: Destroying... [id=i-003c6b1f237f628f7]
aws_instance.server: Still destroying... [id=i-003c6b1f237f628f7, 00m10s elapsed]
aws_instance.server: Still destroying... [id=i-003c6b1f237f628f7, 00m20s elapsed]
aws_instance.server: Still destroying... [id=i-003c6b1f237f628f7, 00m30s elapsed]
aws_instance.server: Destruction complete after 32s
```

Destroy complete! Resources: 1 destroyed.

- Deleted workspace also

```
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace select default
Switched to workspace "default".
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace delete dev
Deleted workspace "dev"!
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace delete staging
Deleted workspace "staging"!
PS C:\Users\LENOVO\Desktop\terraform -workspcae> terraform workspace delete prod
Deleted workspace "prod"!
PS C:\Users\LENOVO\Desktop\terraform -workspcae> 
```